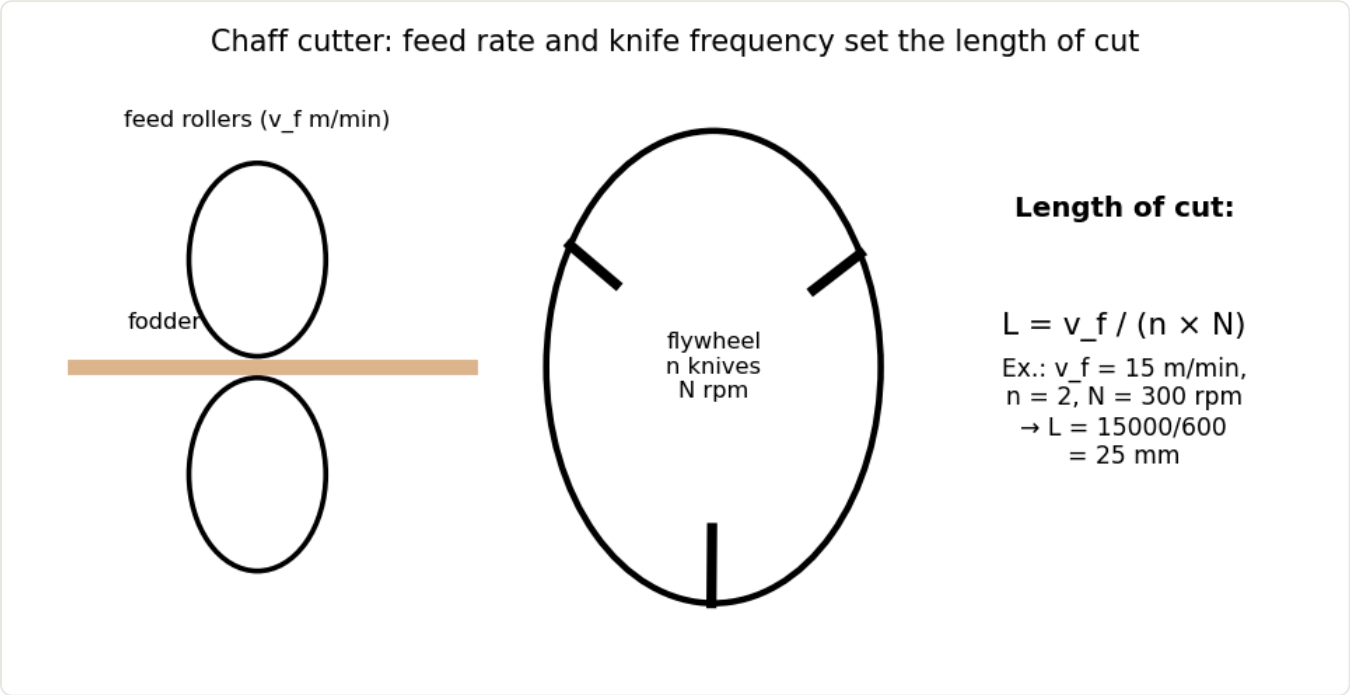
UNIT 3 · TOPIC 7(C)

Chaff Cutters — Length of Cut, Capacity and Losses

Syllabus topic: Calculation of capacity, efficiency and losses in threshers, harvesters and chaff cutters



Chaff cutter — feed rate and knife frequency set the length of cut

Length of cut

$$L = \text{feed velocity} \div \text{cuts per unit time} = v_f/(n \times N)$$

v_f = feed-roller delivery; n = knives on the flywheel; N = flywheel rpm. Adjust L via the feed-roller gearing or the number of knives: finer chaff for cattle (dry roughage ~**10–25 mm**), longer for silage.

> **Worked example.** v_f = 15 m/min, 2 knives, 300 rpm → cuts/min = 2 × 300 = 600 → L = 15,000 mm ÷ 600 = **25 mm**.

Capacity and efficiency

$$\text{Capacity (kg/h)} = \text{throat area} \times \text{material bulk density} \times \text{feed velocity (limited by power)}$$

- Efficiency measures: actual/theoretical capacity, % over-length pieces, specific energy (kWh/t).
- **Safety:** worker-feeding injuries — feeding-chute length per the BIS safety standard, no-kickback rollers (Unit 5).

Practice MCQs

Q95. Chaff cutter length of cut with feed velocity v_f, n knives, flywheel speed N is:

(a) v_f/(nN) (b) nN/v_f
(c) v_f·n·N (d) N/(n·v_f)

Q96. v_f = 15 m/min, 2 knives, 300 rpm → length of cut =

(a) 12.5 mm (b) 25 mm
(c) 50 mm (d) 75 mm

Q97. To get FINER chaff one should:

(a) Reduce knives (b) Slow the flywheel
(c) Reduce feed-roller speed (or add knives) (d) Widen throat

Q98. Chaff cutter throughput capacity is proportional to:

(a) Throat area × bulk density × feed velocity (b) Knife colour
(c) Flywheel weight only (d) Frame length

Answers

Q95 — (a) v_f/(nN)

Q96 — (b) 25 mm — *Cuts/min = 2×300 = 600; L = 15,000 mm/600 = 25 mm.*

Q97 — (c) Reduce feed-roller speed (or add knives)

Q98 — (a) Throat area × bulk density × feed velocity

Sources: Kepner, Bainer & Barger — Principles of Farm Machinery, Ch. 2 (Field Capacities) & Ch. 17; RNAM test codes (combine, thresher, transplanter); BIS standards for power threshers and chaff cutters; Jagdishwar Sahay; NPTEL lectures.